

artificial intelligence (AI). These can also minimise latency, provided the various elements are strategically placed on the motherboard to minimise interference and interconnection delays.

An SoC's building blocks

An SoC combines the power of the CPU with several other components like input and output ports, internal memory, etc to perform and execute its functions. Components that an SoC generally incorporates within itself include:

- A processor or multiple processors at its core to define its functions. It can be a microcontroller (MCU), a microprocessor, a digital signal processor, an application-specific instruction set processor or field programmable gate array (FPGA).
- Internal memories like RAM, ROM, EEPROM or flash to perform computation.
- External interfaces to comply with such industry-standard communication protocols as USB, Ethernet and HDMI. It can also incorporate wireless technology, and involve protocols of Wi-Fi and Bluetooth.
- A graphical processing unit to help visualise the interface.
- Other necessary components such as voltage regulators, phase lock loop control systems and oscillators, clocks and timers, analogue-to-digital and digital-to-analogue converters, etc.
- An internal interface bus or network to connect all individual blocks.

SoC testing

SoCs enable tight integration of components on a single chip, but this necessitates comprehensive testing of electronic devices before dispatch.

Testing of SoC devices has become an increasingly challenging task as these devices have become extremely complex. SoC chips are constructed block-by-block, so testing is efficient when it is also done block-by-block. Designers can install a specialised, configurable, pre-designed embedded system to test and debug each block. Such an embedded system can specify test speed, diagnostic options, fault cov-



An ICT tester (Credit: www.ivp-elec.com)

erage and test length while testing any random logic block.

Compared to CPUs, SoCs use shorter wiring and subsequently utilise less power, making them much more efficient and energy-smart. The only problem with SoCs as compared to CPUs is upgradation and repairing. Where you can easily replace and use new components such as RAM or GPU in CPUs, doing so is a much more complicated process with SoCs.

A few major SoC test equipment manufacturers are Advantest Corp., Astronics Corp., CHROMA ATE Inc., Cohu Inc., Teradyne Inc. and Keysight.

Semiconductor automated test equipment (ATE) consists of various instruments or cards used for testing SoC components, both at wafer and packaged stages. These test systems are continuously evolving as per demands in consumer, computing and communication markets. Today's ATE products must provide more functionality at higher speeds to keep pace with innovation in the semiconductor industry.

Programmable logic plays an indispensable role in ATE product development by providing scalability and flexibility. Such functions as timing accuracy, high-speed I/O capability, digital signal processing (DSP) analysis, memory control and jitter compliance are all taken care of

by programmable logic. With increasing complexity of ATE products, more intellectual property (IP) is persistently being integrated within the programmable logic.

In-circuit testing

Manufacture of electronic assemblies involves many complex tasks that are vulnerable to faults. So, the PCB assembly, components on the board and integrity of the parts are tested to ensure that the end-product meets quality and design standards.

In-circuit testing (ICT) is a widely-recognised, cost-effective method to test medium- or high-volume PCBs. It provides more reliable results as it individually tests PCB components one at a time. ICT assists original equipment manufacturers (OEMs) and contract manufacturers to meet defined parameters consistently during the entire production process.

ICT can be performed with a bed-of-nails test fixture or a flying probe setup because of multiple contact pins PCBs contain. In ICT, test signals are transmitted into and out of PCBs at high speed to evaluate circuits and components. During ICT, all components on the PCBs are measured to detect defective components and replaced accordingly. Function tests check the entire intended function of the PCBs.

Benefits of ICT. An ICT fixture can